

Stellar PSFs Fake Hosts in Little Red Dots

Broad-band point-spread functions built from field stars misassign light when source spectra differ.

arXiv:2608.30459 — Source-conditioned polychromatic PSFs for compact high-redshift nuclei

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Background

Compact high-redshift sources are modelled by subtracting a point-spread function built from nearby field stars. Inferred sizes, luminosities and masses all rest on that subtraction being correct.

Field-star PSFs assume the target's spectrum matches the stars, which little red dots violate.

What this paper is selling

The authors derive why broad-band stellar PSFs systematically misassign light for spectrally distinct compact sources. They demonstrate a correction that removes spurious extended components from little red dots and similar nuclei.

Key results to remember

- A pure point source with no host still yields a spurious 11.8% Sérsic component.
- The error persists across spectra, stellar libraries and all tested detector positions.
- Inferred sizes, luminosities and masses of little red dots are therefore biased.

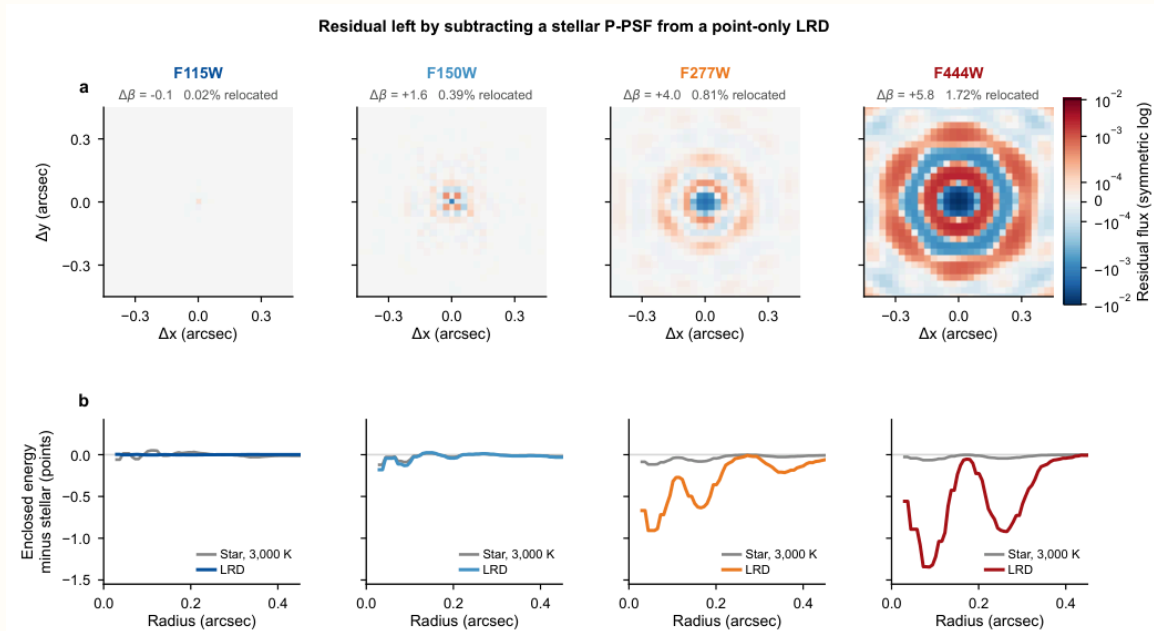


Figure 1 | Subtracting a stellar PSF from a point-only little red dot leaves wavelength-dependent residuals mimicking a Sérsic host.

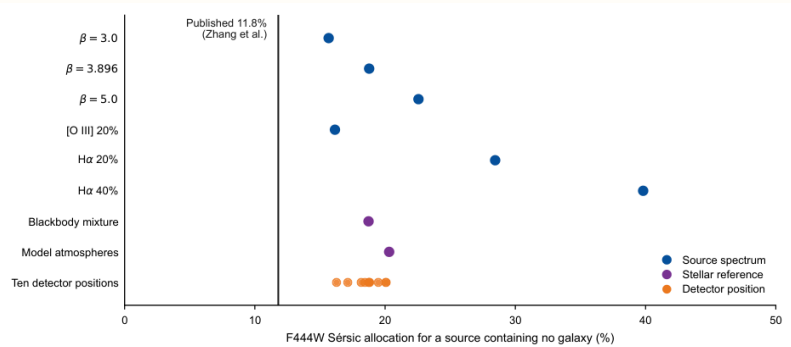


Figure 2 | Every tested nuclear spectrum, stellar reference and detector position allocates more F444W flux than the published 11.8% component.